APPARATUS AND METHOD FOR AN ULTRA-WIDE DATA BUS ARCHITECTURE VIA PARAMETRIC ENCODING

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Title: APPARATUS AND METHOD FOR AN ULTRA-WIDE DATA BUS ARCHITECTURE VIA PARAMETRIC ENCODING

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Cross-Reference to Related Applications

This application claims the benefit of and priority to the portfolio of inventions described in the publication entitled "Be Water, Be Light," the contents of which are hereby incorporated by reference in their entirety. The system described herein is an improvement upon the high-cardinality parallel data bus and parametric series number representation described therein.

Abstract

An apparatus and method for a data bus architecture with a width of multiple billions of bits per clock cycle. The architecture's vast width is achieved not by increasing the physical channel count, but by a novel parametric encoding method that fundamentally changes the nature of the data transmitted. Instead of encoding the value of a number directly,

a single state on a high-cardinality parallel interface is used to encode a multi-parameter "recipe" that describes how to generate a number from a known mathematical series. This method increases the information density of each channel by orders of magnitude, enabling an effective bus width of over 3.2 gigabits and a throughput of over 790 gigabits per second on a commercially available 8K optical display, a width that is over three million times wider than a conventional 1024-bit bus.

Background of the Invention

Prior art, such as that described in the "Be Water, Be Light" portfolio, disclosed a novel physical transmission medium for high-cardinality parallel data transfer, such as an optical interface capable of transmitting millions of discrete states per pixel. This defined a new "medium" for communication, fundamentally different from traditional serial, binary systems. However, the encoding method used with this medium was still based on a one-to-one mapping, where a single state on the interface corresponded to a single number or data value. The total width of the data bus was therefore limited by the number of parallel physical channels (e.g., pixels).

The present invention discloses a novel "language" or encoding method for use with such a medium. While the prior art defined the physical layer (the "medium"), the present invention defines a new logical and data-link layer (the "language") that fundamentally alters the information capacity of the system. This new method breaks the one-to-one correspondence between a transmitted state and a data value, enabling a bus architecture of unprecedented width.

Summary of the Invention

The present invention is an apparatus and method for an ultra-wide data bus that achieves a width of billions of bits by using parametric encoding. The system leverages a high-cardinality parallel interface, but instead of using each state to represent a single number, it uses each state to represent a multi-parameter recipe for generating a number. This recipe may include parameters such as a series identifier, a position within that series, and an offset, or it may be a set of indices into pre-shared matrices of multipliers and series members.

By transmitting a compact recipe instead of a large numerical value, the information density of each channel in the parallel interface is increased exponentially. A single state, representing

a single pixel on an optical display, can now encode the information necessary to reconstruct a number far larger than what could be represented by the base cardinality of the state itself.

When implemented on a commercially available 8K display operating at 240 Hz, this encoding method produces a data bus with a theoretical width of approximately 3.3 gigabits. This is over 3.2 million times wider than a state-of-the-art 1024-bit bus. The invention enables a paradigm shift in computing, moving from architectures defined by physical bus width to architectures defined by the density of the information encoded on the bus.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Federal Government, its military, and its Federal law enforcement agencies a perpetual, royalty-free, non-exclusive license to practice and use the invention described herein for any and all governmental, defense, and non-commercial purposes. This license explicitly excludes any and all state and local governments.

Brief Description of the Drawings

FIG. 1 is a conceptual diagram contrasting the prior art one-to-one encoding method with the parametric "recipe" encoding method of the present invention, illustrating the exponential increase in information density.

FIG. 2 is a diagram providing a comparative visualization of the scale of a conventional 1024-bit bus versus the approximately 3.3-gigabit bus enabled by the present invention.

Detailed Description of the Invention

Referring to **FIG. 1**, a comparison of encoding methods on a single channel (e.g., one pixel) of a high-cardinality interface is shown. In the prior art method (100), the state of the pixel (101) is mapped directly to a single numerical value (102). The information content is limited by the base cardinality of the interface. In the present invention (110), the state of the pixel (111) is mapped to a parametric recipe (112), which contains multiple pieces of information (e.g., Series ID, Position, Indices). A decoding module (113) uses this compact

recipe to reconstruct a potentially massive numerical value (114). The information content is thus exponentially greater than the base cardinality of the interface.
Referring to **FIG. 2**, the immense scale difference in bus width is visualized. A conventional 1024-bit bus (201) is shown as a very thin bar. In contrast, the ultra-wide parametric bus of the present invention (202) is shown as a massively larger bar. A label (203) indicates that the new bus is over "3.2 Million Times Wider," providing a clear, immediate understanding of the revolutionary scale of the invention. The bus width of the present invention, when implemented on an 8K display, is approximately 3,306,598,656 bits (204).

Claims

What is claimed is:

1. A method for transmitting data across a parallel interface, comprising the steps of:
    - providing a parallel interface comprising a plurality of channels, each channel capable of representing one of a set of discrete states;
    - encoding a numerical value as a set of a plurality of parameters, wherein said parameters provide instructions to generate the numerical value from a mathematical series; and
    - mapping the entire set of said parameters to a single discrete state of a single channel of the parallel interface, thereby creating an effective data bus width that is a multiple of the number of physical channels.
2. The method of claim 1, wherein the effective data bus width is greater than 1,000,000 bits.
3. The method of claim 1, wherein the effective data bus width is greater than 1 gigabit.
4. The method of claim 1, wherein the set of parameters comprises a series identifier and a position within the series.
5. The method of claim 1, wherein the parallel interface is an optical interface and each channel is a pixel.
6. An apparatus for data communication, comprising:
    - a parallel data bus comprising a plurality of physical channels, each capable of representing a discrete state from a high-cardinality set of states;

- an encoding module configured to represent a numerical value as a single data word, wherein said data word comprises a set of parameters for generating the numerical value from a mathematical series; and
- wherein the encoding module is configured to map said data word to a single discrete state of a single physical channel.

7. The apparatus of claim 6, wherein the apparatus is configured to operate with an effective bus width of over 3.2 million times the width of a 1024-bit bus.

8. The apparatus of claim 6, wherein the parallel data bus is an 8K resolution optical display.

Drawings

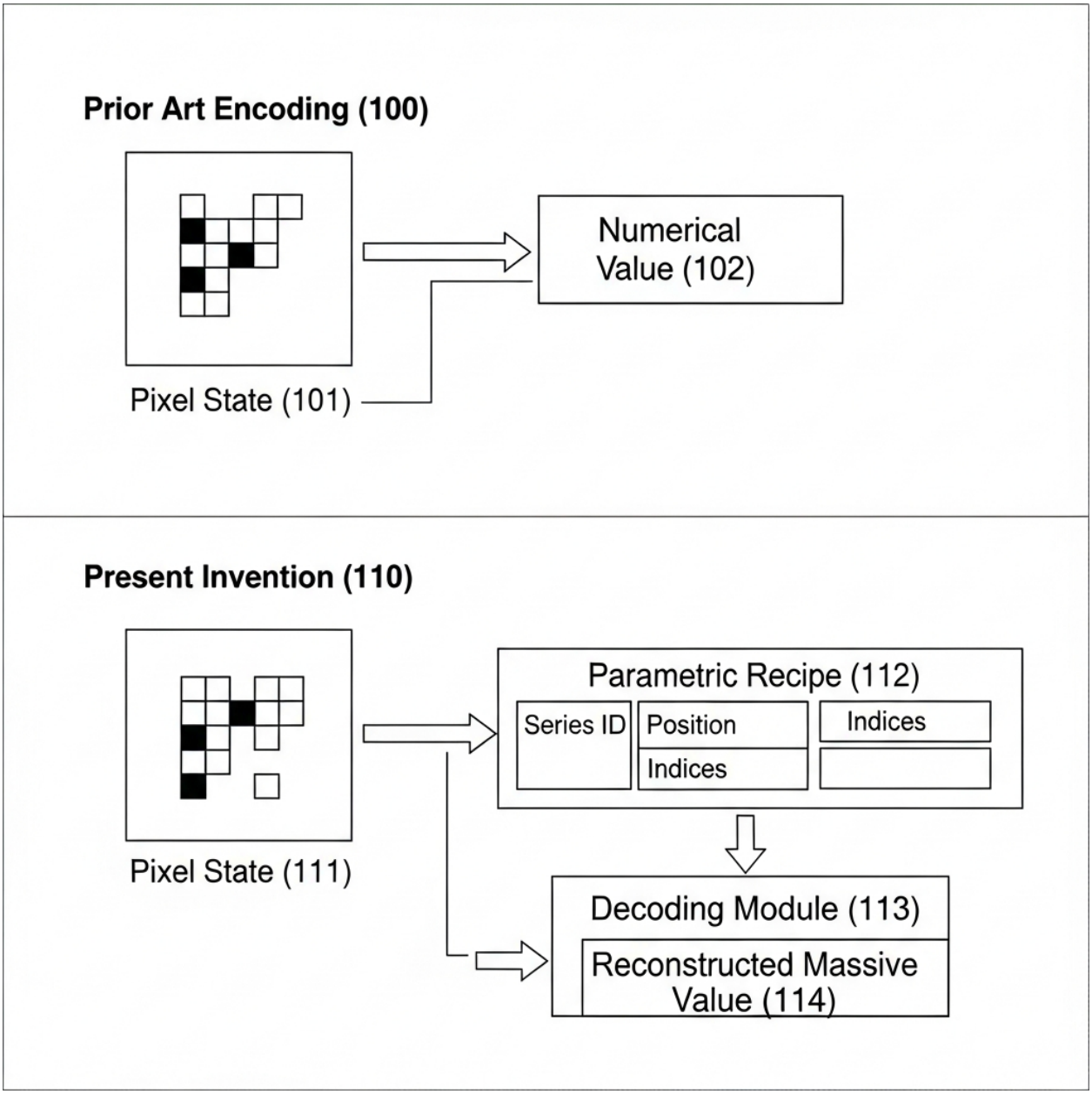


Figure 1: Conceptual diagram contrasting one-to-one vs. parametric encoding.

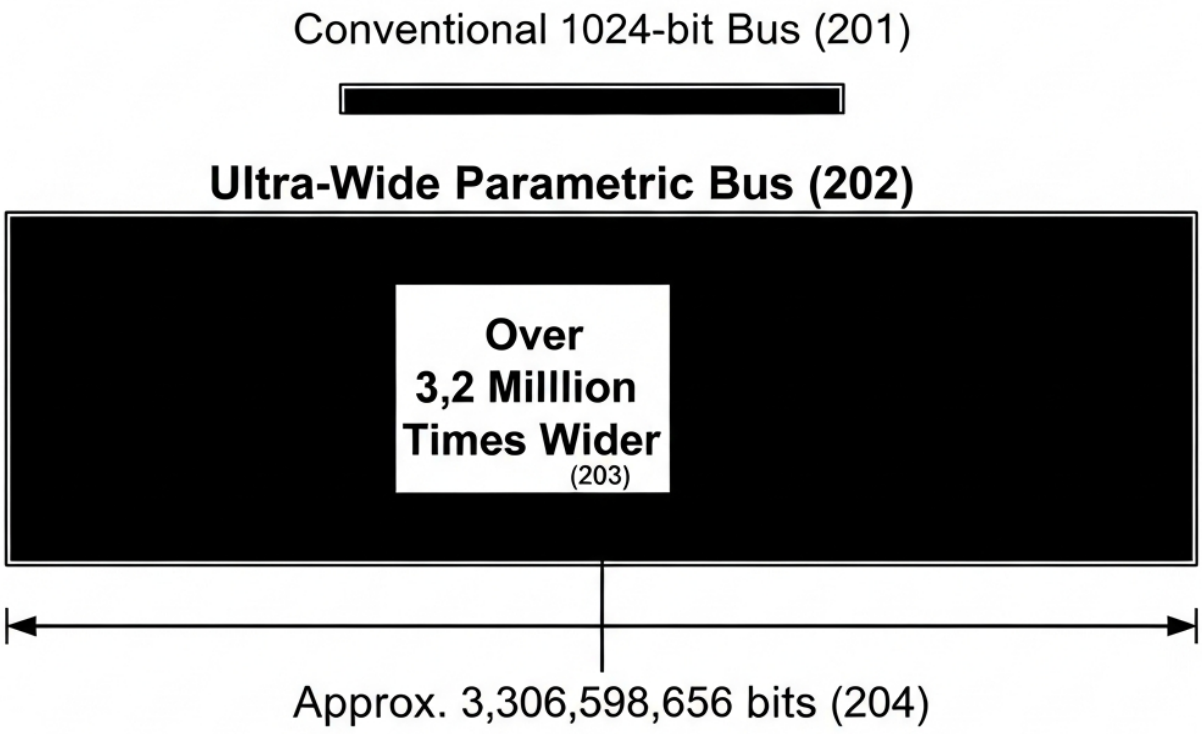


Figure 2: Diagram visualizing the comparative scale of the bus width.